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Research on Evaluation Methods of Complex Product Design Based on Hybrid Kansei Engineering Modeling

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Abstract: The field of complex product design evaluation can attract high ambiguity due to difficulties in establishing indicators and the subjectivity of expert evaluation scoring. Indeed, traditional Kansei Engineering (KE) relies on user requirements and feedback for design evaluation, which may not fully and effectively validate the design evaluation results, let alone determine whether they apply to complex products with more evaluation index systems. To overcome these drawbacks, this study proposes an evaluation method based on Hybrid Kansei Engineering (HKE) modeling for complex product design evaluation. HKE modeling consists of two parts, namely Forward Kansei Engineering (FKE) and Backward Kansei Engineering (BKE). In this study, four electric forklift designs are used as an example. The FKE system adopts the multi-attribute decision evaluation method; obtains the evaluation indexes of the forklift product imagery and then establishes the perceptual word collection; constructs the evaluation index system of the forklift via the Analytic Hierarchy Process (AHP); calculates the weights of the evaluation indexes of each level and their rankings; and calculates the final rankings of the four electric forklift design solutions by adopting the Fuzzy Comprehensive Evaluation (FCE) method. The BKE system adopts eye tracking (ET) to extract the attention time, visual attention hotspot, and other eye movement index data, and the Gray Relation Analysis (GRA) method was used to validate the model to derive the ranking, which verifies the effectiveness and scientific validity of the evaluation method. The results of this study show that the two-way evaluation of HKE modeling can effectively avoid subjective factors in product design, improve the scientific nature of the design, and guarantee the logical rigor of the perceptual design procedure.

Keywords: product design evaluation; hybrid Kansei engineering; complex product design; fuzzy comprehensive evaluation; eye tracking

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1. Introduction

Product design evaluation is a complex and fuzzy decision-making process due to the influence of culture, values, psychological activities, and other aspects, and user needs are typically general and vague [1]. Previously, product designers used their experience and design principles to complete a design scheme and then evaluated the scheme through methods such as expert scoring forms or user demand feedback research [2]. Although such a design evaluation process is highly efficient, it is also highly subjective, and the design solutions obtained are largely dependent on the designer's experience and aesthetics. The externalization of user requirements and the combination of quantitative research methods

for comprehensive and accurate program design evaluation are inevitable trends in the development of industrial design [3].

Complex products cover the basic requirements of traditional product design and have the characteristics of high design cost, large scale, many sub-modules, high technical requirements, and complex design indexes, which leads to the design process not fully using the traditional design methods of large-scale product manufacturing [4–6]. Complex product design evaluation is a multi-task, multi-layer, uncertain, and complex decision-making process [7]. Currently, there is a lack of effective tools and methodologies to support the design of complex products. It is therefore necessary to conduct a comprehensive assessment of the design scheme based on design specification experience and other relevant data [8] so that problems that may arise during product use can be identified and solved in the design phase, thus reducing product costs, shortening the development cycle, and improving product competitiveness.

Based on the background outlined above, this study explores three research questions (RQs): RQ1: In the current context of over-reliance on designers' subjective feelings in product design evaluation, is there an evaluation methodology superior to traditional KE that can quantify user requirements and accurately evaluate complex product design solutions? RQ2: How can we minimize subjectivity in the evaluation of complex product designs such as electric forklifts, and what methods are effective in quantifying user requirements during the expert subjective evaluation process? RQ3: Which experiment is employed to accurately validate the evaluation results? How can experiments be designed and ensure the validity and rationality of the evaluation?

To address the questions above, this study proposes a complex product design evaluation method based on Hybrid Kansei Engineering (HKE) modeling, in which forward HKE adopts the evaluation method of the Analytic Hierarchy Process (AHP) and Fuzzy Comprehensive Evaluation (FCE) to obtain the design solution ranking and reverse HKE verifies the evaluation result via an eye-tracking (ET) experiment. This study thus makes the following contributions to the literature.

To address RQ1, this study adopts HKE modeling and applies it to solve the complex product design evaluation problem. The reliability and consistency of the two sets of evaluation data are examined by comparing the forward and the reverse evaluation results. HKE has the characteristic of two-way translation, which has an advantage over traditional KE in terms of design decision accuracy, and is suitable for the evaluation of more complex product designs. This study demonstrated the validity of the results through the validation of an electric forklift.

RQ2 illustrates the importance of an effective decision evaluation model for product design. This study combines AHP and FCE methods, which can facilitate more comprehensive and accurate product evaluation decisions, provide key guidance for multi-attribute decision evaluation problems, and make product evaluation decisions more reliable.

To address RQ3, this study adopts the ET technique. The user's preference is obtained by analyzing the eye movement index data, and the design solution is evaluated and validated via Gray Relational Analysis (GRA). The root mean square error (RMSE) is used to scientifically measure the output data result validity of subjective and objective assessment methods. The validity and rationality of the theoretical modeling mechanism in this study are then verified again via subjective and objective methods.

The rest of this manuscript is structured as follows. Section 2 provides related research work on product design evaluation methods to find the entry point of this study. Section 3 explores the research methodology adopted in this study, including HKE modeling, the Fuzzy AHP evaluation method, and an experiment using the eye-tracking technique. A complex product design evaluation method based on HKE modeling is constructed, and

the steps of the proposed method are described. Section 4 takes an electric forklift (HELI, China) as an example to prove the validity and feasibility of this study. Finally, Section 5 summarizes the study and gives suggestions for future research directions.

2. Literature Review

To date, a wide range of research on product design evaluation using Kansei Engineering (KE) modeling has been widely developed. Sections 2.1 and 2.2 review the literature on Kansei Engineering and product design evaluation methods, respectively, and Section 2.3 reviews design evaluation studies for complex products and gives a comparison and summary of related literature.

2.1. Kansei Engineering Model

KE is a theory based on design science, psychology, cognition, and other related disciplines that introduces human perceptual analysis into the field of engineering technology [9]. It can transform perceptual user needs into product design elements, and in the field of product design, KE is a user-oriented product research and development technology [10]. The correct application of perceptual information can stimulate designers to acquire and express users' design needs and elements, and can also help users find their favorite products from a large number of similar products [11]. Perceptual engineering has therefore been the focus of many design scholars in recent years.

However, in recent years, research articles have found that studies on KE have introduced perceptual imagery into the product design process, extracted perceptual design elements, and guided the output of the design solution, with no detailed inference when evaluating the decision [12]. Undoubtedly, this traditional approach provides efficient results, but there are some drawbacks, such as the inability to verify the accuracy between perceptual needs and final design solutions [13]. Designers and users are willing to use perceptual engineering to solve product design problems, but deeper rethinking is needed at the methodological level to ensure that perceptual needs and product attributes are obtained along with accurate product design solutions.

There have been several studies expanding on methods based on KE. In response to the difficulty of KE in dealing with the ambiguity of user requirements, Fuzzy Kansei Engineering (FKE) has been more widely applied by researchers. HuiCun Shen. et al. [14] used FKE to quantifiably explore present user preferences and their relationship with product design elements. Akgül. et al. [15] found that fuzzy language can effectively deal with the vagueness of user requirement data in KE and extract user requirement data via fuzzy rules in a more comprehensible structure. Lei Fu. et al. [16] proposed an innovative integrated approach model to express users' opinions on product design evaluation by combining hesitant fuzzy numbers with AHP and QFD methods, improving the accuracy of user requirement acquisition in FKE. Collaborative Kansei Engineering (CKE) improved the accuracy of design by gathering the requirements of multiple groups, including designers, users, and stakeholders, and engaging them in guiding the design. Zhou Tie. et al. [17] developed a new intelligent wheelchair design system by collecting the life requirements of the elderly, disabled, and other groups, which can effectively improve and enhance the quality of daily life of users. Virtual Kansei Engineering (VKE) mainly used VR technology to study the correlation between product design elements and user requirements. Qianwen Fu. et al. [18] adopted a research method that combines the VR system interface with KE to analyze user satisfaction and constructed a spatial perception model of user images. Xing Liu et al. [19] proposed a product design method based on VR and KE, demonstrating that VR is more suitable for measuring and predicting users' reactions to future products. HKE improved the accuracy of product design evaluation by using a combination of

forward evaluation and reverse validation. Kun-Chieh Wang [20] explored the relationship between user requirements and product design using Gray Systems Theory (GST) and Support Vector Regression (SVR) in HKE. This paper explored the design evaluation of complex products based on the HKE model, and a combination of FCE, AHP, FCE, ET, and GRA methods were used in the forward assessment and reverse validation phases to demonstrate scientific validity and accuracy. A comparison of extension methods based on KE is shown in Table 1. In summary, FKE and CKE improved the accuracy of design from user requirement acquisition. VKE and HKE focused on combining other techniques or methods to solve product design problems, with the difference that VKE relies mainly on virtual technology, while HKE does not limit the combination with any method or technique.

Table 1. Comparison of extension methods based on Kansei Engineering (KE).

Kansei Engineering Extension Method	Highlighting Advantages	Author and Reference	Year	Research Contribution
Fuzzy Kansei Engineering	Resolved ambiguity of user requirements	HuiCun Shen. et al. [14]	2016	Explored user preferences and their relationship with product design elements.
		Akgül. et al. [15]	2022	Found that fuzzy language can deal with the vagueness of user requirement data in KE.
		Lei Fu. et al. [16]	2024	Proposed an innovative integrated methodology model incorporating HFAHP and HFQFD.
Collaborative Kansei Engineering (CKE)	Improved the accuracy of demand acquisition through deep involvement of multiple groups	Zhou Tie. et al. [17]	2010	Developed a new intelligent wheelchair design system by collecting the life requirements of the elderly, disabled, and other groups.
Virtual Kansei Engineering (VKE)	Integrated VR technology to study correlation between product design and user needs	Qianwen Fu. et al. [18]	2020	Adopted a method that combines the VR system interface with KE to analyze user satisfaction.
		Xing Liu et al. [19]	2022	Proposed a product design method based on VR and KE, demonstrating that VR is more suitable for measuring and predicting users' reactions.
Hybrid Kansei Engineering (HKE)	Improved accuracy of product design evaluation by combining forward evaluation and reverse validation	Kun-Chieh Wang [20]	2011	Explored the relationship between user requirements and product design using GST and SVR.
		This paper		Explored the design evaluation of complex products based on HKE, and a combination of FCE, AHP, FCE, ET, and GRA was used in the forward assessment and reverse validation.

2.2. Product Design Evaluation Methods

Product design evaluation methods mainly include subjective evaluation, objective evaluation, comprehensive evaluation, and experimental research [21]. Subjective evaluation relies on designer experience to elicit judgment, for example, via the expert interview method. The objective evaluation method is based on data [22], and the conclusion is calculated via quantitative evaluation methods. Methods such as the AHP, utility theory, Quality Function Deployment (QFD), Technique for Order Preference by Similarity to an Ideal

Solution (TOPSIS), graphical tools, and fuzzy logic are the most popular in product design evaluation [23–25]. A comprehensive evaluation combines multiple evaluation methods, including subjective and objective evaluation, which can simultaneously eliminate the effects of subjectivity and uncertainty [26]. A comprehensive evaluation is therefore suitable for more complex product evaluation objects. The experimental research method is applied to establish causal relationships between variables through experimental operations and to assess the effect it has on other indicators through the change indicators of one or more variables. In product design, experiments are mainly conducted on a certain physiological index of the user, including surface electromyography (sEMG), electroencephalography (EEG), and the ET method [27,28], among which ET is the most commonly used because it is directly related to vision and can intuitively respond to a user's preferences [29].

2.3. Complex Product Design Evaluation

Complex products have more evaluation indexes and fuzzy evaluation information, and the evaluation process needs to be discussed and compared with multiple methods [30]. Currently, with developments in quantitative evaluation methods and improvements in product design requirements, research literature on design evaluation methods for complex products is fast emerging. Yuan Chang et al. [31] developed a smart industrial service ecosystem in a complex industrial context, and achieved it by combining the Fuzzy Kano and AHP methods in a Pythagorean fuzzy (PF) environment. Xinhui Kang [32] used support vector regression to establish a mapping model between customer perceptions and key morphological characteristics, and obtained the optimal product design solution with the highest perceptual value. Wang Shuai et al. [33] proposed a hybrid structure decomposition method to obtain all components of complex products according to the different structural characteristics of each complex product subsystem. Zhe Chen et al. [34] proposed a novel integrated two-layer method for product design to determine expert weight under these circumstances. Raffaeli et al. [35] proposed a framework to support and guide a designer by applying a structured and algorithmic procedure to evaluate the opportunity to adopt Additive Manufacturing and achieve an optimal design. Arbabi et al. [36] developed a quantitative data-driven Multi-Criteria Decision-Making (MCDM) approach, founded on the combination of an artificial neural network and TOPSIS, to assess the new conceptual designs. Zhu et al. [37] proposed an approach that integrates electroencephalography (EEG) and eye-tracking (ET) data in a new way to derive multi-faceted supportive information for product evaluation. Zhu et al. [38] built a design research model based on AHP, QFD, and PUGH decision matrices, effectively avoiding subjective factors in product design. Yong Wang et al. [39] proposed a compensatory evaluation method for color schemes based on eye movement tracking technology to conduct a more objective quantitative evaluation for color schemes. An overall comparison between this study and previous studies is shown in Table 2. Complex products examined in previous studies include engineering instruments, automotive vessels, medical devices, and energy parks, which require more comprehensive and accurate evaluation methods than do conventional products. This study is based on HKE modeling for an electric forklift, and the methods used include AHP, FCE, ET, and GRA.

Table 2. Overall comparison between this study and previous studies.

Author and Reference	Complex Product	Case Evaluation Methodology	Results Comparison Validation Method	Research Contribution
This study	Electric forklift	HKE, AHP, FCE	ET, GRA	Proposed a complex product design evaluation method based on hybrid perceptual engineering.
Yuan Chang et al. [31]	Renewable energy park	PF-AHP, PF-Kano	F-AHP, F-Kano	Developed a smart industrial service ecosystem in a complex industrial context.
Xinhui Kang. [32]	Electric vehicle	KE, RST, SVR		Explored the mapping relationship between customer visual sensibility and key morphological characteristics based on KE.
Wang Shuai et al. [33]	Hydraulic excavator	DSM	Modularity evaluation	Presented a core component-oriented modularity design method for complex products.
Zhe Chen et al. [34]	Cruise ship cabin	EW, Multiplicative AHP	Minimized variance model	Presented an integrated expert weight determination method for design evaluation.
Raffaeli, Roberto et al. [35]	Ultralight aircraft part	WSM, AHP		Proposed a framework to support and guide a designer by means of a structured and algorithmic procedure for the adoption of AM.
Arbabi, Hamidreza et al. [36]	Automotive design	MCDM, TOPSIS		Developed a data-driven MCDM approach to assess conceptual automotive designs.
Zhu, Siyu et al. [37]	Automotive interior	STNN, fuzzy system	EEG, ET	Presented an approach for automotive interior evaluation based on integration of EEG and ET.
Zhu, T. L. et al. [38]	Surgical auxiliary equipment	AHP, QFD, PUGH		Built a design research model for medical equipment based on AHP, QFD, and PUGH decision matrices.
Yong Wang et al. [39]	Military jungle camouflage	Compensatory evaluation	ET	Proposed a compensatory evaluation method for a color scheme based on ET.

Note: PF-AHP: Pythagorean fuzzy Analytic Hierarchy Process; KE: Kansei Engineering; RST: rough set theory; SVR: support vector regression; DSM: design structure matrix; EW: entropy weight method; WSM: Weighted Sum Model; MCDM: Multi-Criteria Decision-Making; TOPSIS: Technique for Order Preference by Similarity to an Ideal Solution; STNN: spatial–temporal neural network; EEG: electroencephalography; QFD: Quality Function Deployment.

3. Materials and Methods

3.1. HKE Modeling

Because subjective human feelings are often very complex and might conflict with each other, product design evaluations based on traditional KE show lower-quality results in terms of generalizability [40,41]. From this basis, the concept of HKE was first proposed by Matsubara and Nagamachi [42]. The HKE model is shown in Figure 1, and consists of two sub-systems, namely FKE and BKE. In FKE, user requirements are inputted through perceptual imagery vocabulary, and design options and design evaluations are outputted;

in BKE, design elements or design options are inputted, and design strategies and design evaluations are outputted.

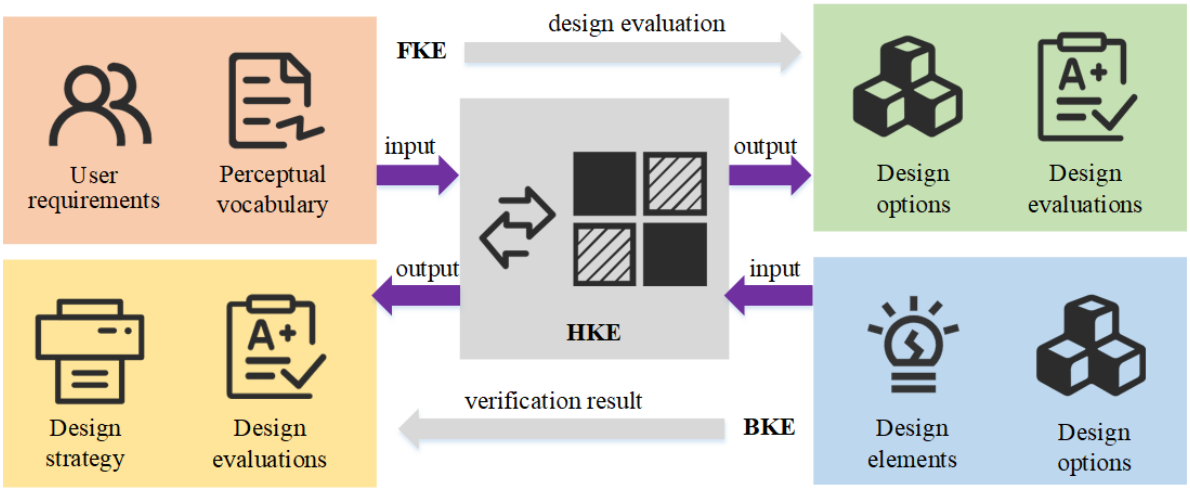


Figure 1. Overview of the proposed HKE model.

Compared to traditional KE, HKE is more comprehensive and accurate in guiding the product design evaluation process. KE lacks a robust validation mechanism and effective evaluation methods for product design. Although KE can effectively capture user emotions and needs, its data collection and processing methods primarily rely on expert opinions or user surveys, with no effective means to validate whether these subjective evaluations accurately reflect the design aspects of the product. As a result, traditional KE methods often fail to ensure the reliability and effectiveness of design evaluation results. In contrast, HKE addresses this limitation by integrating both subjective and objective evaluation methods. HKE covers all stages of the design evaluation process, from design to assessment, with the design being verified through evaluation. Specifically, FKE is used to evaluate the design after the design solution is obtained, while BKE revalidates the evaluation results through other quantitative evaluation methods. FKE typically starts from user needs, guiding design through emotional evaluations and user feedback. In the FKE phase, the design team determines the design direction and evaluation focus by collecting user needs through methods such as user surveys, expert interviews, and focus group discussions. BKE, on the other hand, is the validation phase of the HKE model, where quantitative methods or experimental techniques are used to verify whether the design evaluation scheme is scientifically and reasonably assessed. This two-way validation mechanism ensures that the evaluation results are both objective and practical, improving the accuracy of the design process. While traditional KE may be more suitable for simpler products, HKE modeling integrating FKE and BKE provides more objective and accurate evaluation results, making it better suited for evaluating the design of complex products or systems.

The design process of HKE spans from the initial identification of the design object to the final evaluation and validation of the design. This process is more comprehensive and complex, involving multiple stages and the integration of various methods. Figure 2 illustrates the initial flowchart of the HKE design process. Firstly, the design object needs to be identified, and samples and perceptual vocabularies related to the product need to be collected separately. After analyzing and filtering, a perceptual vocabulary library and a sample library are established, and the two libraries are correlated to extract the key elements of the product design. Secondly, product design solutions are obtained based on the design elements, and appropriate methods are selected to evaluate these

design solutions in order to obtain preliminary evaluation results. Next, the preliminary evaluation results are validated by selecting different product evaluation methods to ensure the reliability of the evaluation results. Finally, after comparing and discussing the validated results, the final ranking of design options and the preferred option are obtained based on the validation results.

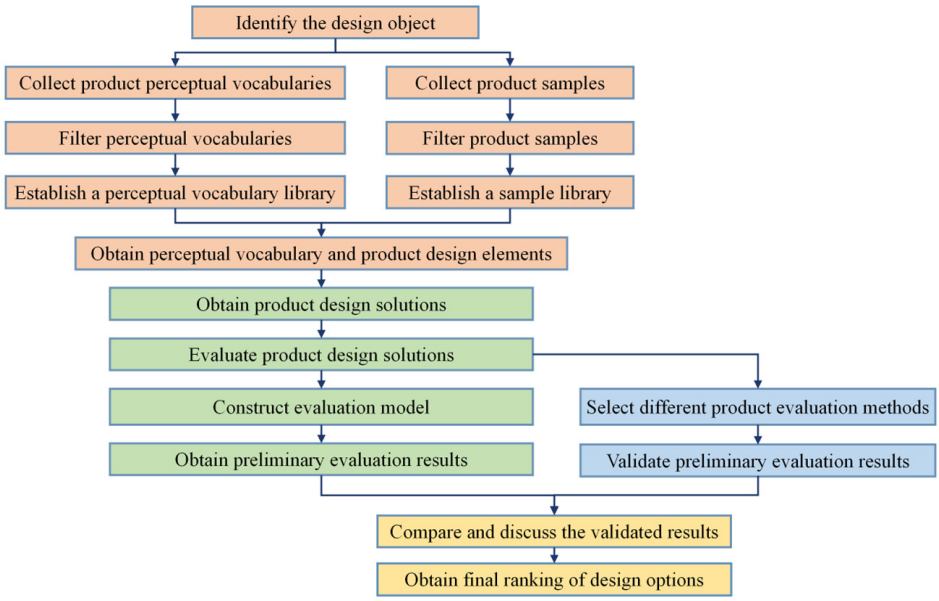


Figure 2. The flowchart of the HKE design process.

In addition, there are two additional improvements to HKE in this study. The first point is that this study gives a range of research objects suitable for applying HKE modeling. This paper extends it to the evaluation of product designs of higher complexity, which enhances the scope and applicability of the HKE model in modern industrial design. Specifically, this paper applies HKE to the evaluation of complex product designs, which not only involve appearance design but also include multiple levels of design criteria and more complex user needs. HKE can precisely meet such higher requirements through two-way evaluation and verification. Therefore, the application of HKE to focus on complex product design evaluation is very appropriate. The second point is that in this paper, HKE modeling incorporates multiple dimensions of qualitative methods, quantitative evaluation methods, and experimental methods, specifically including the introduction of PCA, AHP, FCE, ET and GRA methods, which combine to significantly improve the scientific validity and reliability of the design evaluation process.

3.2. FKE for Design Evaluation Using AHP and FCE Methods

Given the scientific validity and practicality of the design evaluation results, the AHP and FCE methods are used in the FKE model. AHP is a decision support research method proposed by the American academic T. L. Saaty, professor of operations research, and is widely used in various fields related to decision making [43–45]. In a system constructed via AHP, it is necessary to calculate the relative importance of the interrelated factors layer by layer, compare them in pairs, and form a judgment matrix as the basis for the calculation and analysis [46]. The judgment matrix used in this study is composed of the classical method 1–9 scale in hierarchical analysis; the scale values of the judgment matrix are shown in Table 3.

Table 3. Judgment matrix scale.

Scale (m and n)	Importance	Scale Description
1	Equally important	Metric m and n are equally important
3	Slightly important	Metric m is slightly more important than metric n
5	Clearly important	Metric m is clearly more important than metric n
7	Strongly important	Metric m is strongly more important than metric n
9	Absolutely important	Metric m is absolutely more important than metric n
2, 4, 6, 8	Median	Compromise

The judgment matrix is constructed by comparing the elements of the evaluation indicators at each level. The judgment matrix (1) is obtained through a two-by-two comparison of the secondary evaluation indicators:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \quad (1)$$

The hierarchical weight vector summation formula is solved as follows:

$$ai = \sum_{k=1}^{n} eik(i = 1, 2, \ldots n) \quad (2)$$

The elements of each column of the judgment matrix are normalized, and the calculation formula is as follows:

$$aij = \frac{aij}{\sum_{k=1}^{n} akj}(i = 1, 2, \ldots, n; j = 1, 2, \ldots, n) \quad (3)$$

$$\omega i = \frac{\omega i}{\sum_{j=1}^{n} \omega j}(i = 1, 2, \ldots, n; j = 1, 2, \ldots, n) \quad (4)$$

After solving the judgment matrix and weights for each element, a consistency test is performed, calculated as follows:

$$CI = (\lambda max - n)/n - 1 \quad (5)$$

$$CR = CI/RI \quad (6)$$

The smaller the *CI* value, the better the consistency. Generally, $CI < 0.1$ means that the consistency requirements are met.

Calculate the weight of the evaluation results of the criterion layer for design option A, as follows:

$$Pn = \omega n \times Rn \quad (7)$$

Based on the results of the calculations, a second-level comprehensive evaluation matrix was created and the results were converted into a percentage system. The scores of the options in each group were calculated sequentially, and the results of the FKE design evaluation using AHP and FCE were obtained by ranking the scores.

3.3. BKE for Validating Evaluation Results Using the ET Method

In establishing the BKE model, ET technology and GRA are integrated for subjective and objective program evaluation. To efficiently and accurately obtain product design evaluation results, the comprehensive evaluation method combined subjective and objective evaluation. In subjective evaluation, the scoring results of each expert were obtained through expert scoring to capture user needs and emotional preferences; in objective evaluation, eye-tracking and gray correlation analysis techniques were used to collect and analyze actual user behavior data.

This study chose eye-tracking experiments primarily because eye tracking effectively reveals users' attention distribution and emotional responses during product usage, which was crucial for understanding users' design preferences. Eye-tracking technology monitored the movement trajectory of users' gaze in real time, accurately reflecting their attention to different design elements during interaction. Compared to EEG and sEMG, eye tracking had a more direct and clear advantage in capturing visual attention, enabling researchers to better understand users' emotional reactions and preferences when faced with design proposals, thereby providing more valuable insights for product design evaluation. ET can capture the visual movement patterns of the human eyeball and facilitate a deeper understanding of psychological changes in users, which has become an effective means of exploring the perceptual requirements and cognitive preferences of users [47]. The BKE model takes the attention duration in the interest area as the main observation index to determine the user's interest in the design solution. After completing the eye movement test, the subjects were asked to perform a subjective preference test, and the test data were subjected to a gray correlation analysis. The calculation process includes solving the gray correlation coefficient $\delta_i(k)$ and calculating the gray correlation degree $\gamma\left(C_o', C_i'\right)$:

$$\delta_i(k) = \frac{\Delta_{min} + \rho\Delta_{max}}{\Delta_i(k) + \rho\Delta_{max}}, \rho \in (0,1) \tag{8}$$

where ρ denotes the resolution factor, which is usually taken as 0.5.

$$\gamma\left(C_o', C_i'\right) = \frac{1}{n}\sum_{i=1}^{n} \delta_i(k), k = 1, 2, \cdots, m \tag{9}$$

To scientifically measure the validity of the output data results of the subjective and objective assessment methods, the root mean square error (RMSE) was used to calculate the following:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(t_i - x_o)^2}{n}} \tag{10}$$

where n denotes the number of target items; t_i denotes the attention time within the AOI of the design scheme; x_o denotes the gray correlation of each design solution; and $x_o = \gamma\left(C_o', C_i'\right)$. The closer the data are to 0, the more credible the data are.

3.4. An Overview of the Proposed Framework

This study proposes a systematic HKE-based method for complex product design evaluation. In Figure 3, the detailed methodology of the research framework is divided into three parts.

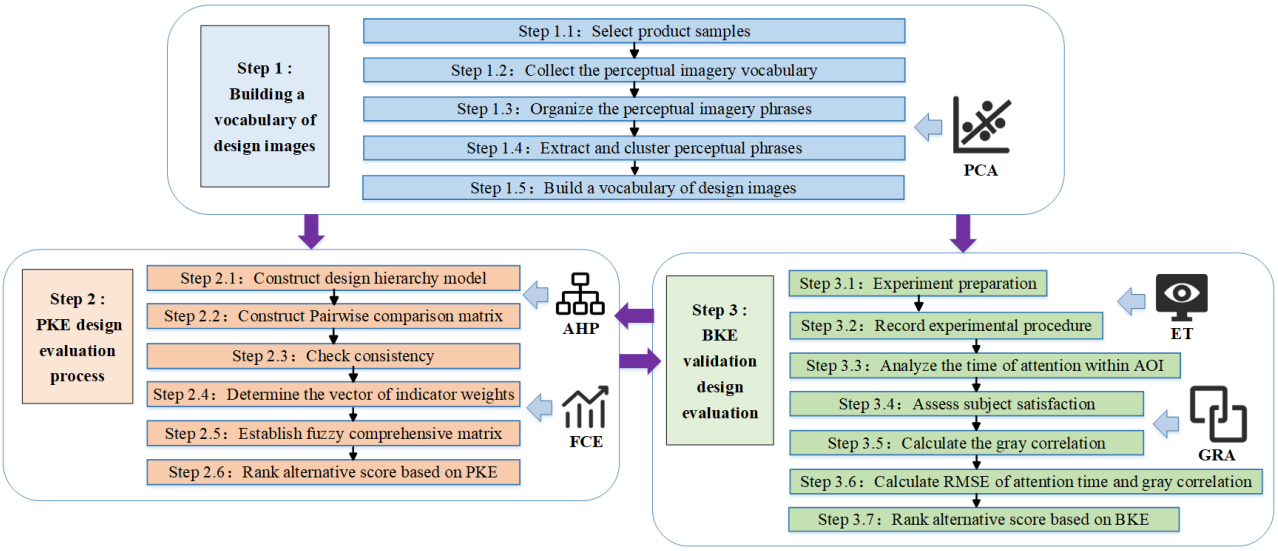


Figure 3. Proposed methodology for complex product design evaluation.

4. Case Study

This section presents a case study to demonstrate the application of the proposed method. The HeLi Corporation (Hefei, China) is a large-scale vehicle research and development company based in China, showing strong economic performance. We therefore describe four electric forklifts from the HeLi Corporation as the research objects to verify the feasibility of the method.

4.1. Building a Vocabulary of Design Images

Given that the design differences in electric forklift trucks are caused by functional factors such as load tonnage, this study selected the widely used 1–3.5 ton battery counterbalanced forklift trucks as the object of study. This study selected electric forklifts as the research subject primarily due to their widespread applications and complex design requirements. As a highly complex industrial product, the electric forklift plays an important role in various industries such as logistics, manufacturing, construction, and warehousing. Its design directly affects production efficiency, work safety, and cost control in enterprises, making the electric forklift an essential and representative piece of equipment in modern industry. However, despite its significant practical implications, academic research on the design of electric forklifts remains relatively scarce, particularly in the areas of user needs analysis and design evaluation decisions, where there exists a substantial gap. Therefore, choosing the electric forklift as the subject of this research not only helped fill the research gap in this field but also provided valuable practical case studies and theoretical support for the design evaluation methods of complex products.

A total of four products with the highest utilization rate and that meet the design requirements were selected; the scheme of the four electric forklifts is shown in Figure 4.



(**a**) Option A (**b**) Option B (**c**) Option C (**d**) Option D

Figure 4. The scheme of the four electric forklifts.

Perceptual imagery vocabulary (Table S1 in Supplementary Materials) was collected using engineering apparatus data and interviews with internal personnel of the enterprise, and a total of 96 perceptual phrases were collected. After discussions and screening by ten experts, involving five industry experts and five designers, the KJ method was used to eliminate words with little relevance to electric forklift design. A focus group discussion was used to screen the words according to the requirements set by the principle of "secondary nature" of the semantic difference method, and 60 perceptual words were finally determined. They were organized into 30 sets of perceptual imagery phrases for electric forklift design evaluation, as shown in Table 4. Participation in the focus groups was voluntary, and the experts were fully informed of the objectives, procedures, and rights of the study, and they gave informed consent prior to participation.

Table 4. Total perceptual imagery phrases for electric forklift design evaluation.

Number	Imagery Phrases	Number	Imagery Phrases
1	Monotonous–Creative	16	Integral–Fragmented
2	Dynamic–Stable	17	Practical–Useless
3	Harmonized–Dissonant	18	Symmetrical–Arbitrary
4	Comfortable–Uncomfortable	19	Safe–Dangerous
5	Lightweight–Heavy	20	Modern–Old-fashioned
6	Tough–Soft	21	Fine–Rough
7	Bright–Dark	22	Durable–Fragile
8	Low-profile–High-profile	23	Exaggerated–Plain
9	Technological–Traditional	24	Energy efficient–Energy consuming
10	Beautiful–Ugly	25	Intelligent–Manual
11	Simple–Complex	26	Conservative–Innovative
12	Strong–Weak	27	Metal–Plastic
13	Organized–Disorganized	28	Expensive–Inexpensive
14	Bright–Gray	29	Efficient–Inefficient
15	Superior–Inferior	30	Reasonable–Unreasonable

The four forklift design cases were scored based on 30 sets of imagery phrases. A five-level Likert scale questionnaire for electric forklift program design (Questionnaire S1 in Supplementary Materials) was produced according to the following five-level evaluation criteria: very important, important, generally important, unimportant, and very unimportant, scored as 2, 1, 0, −1, and −2, respectively. Fourteen participants were invited to score the design image of the electric forklift, including four company employees, two teachers, and eight graduate students, ensuring the diversity of the sample. Given consideration of time and cost constraints, this sample size provided effective evaluation results. The average score value for each perceptual imagery phrase was calculated, as shown in Table 5.

Prior to performing PCA, the data were standardized, setting the mean of each variable to zero and the standard deviation to one. PCA was then applied to reduce the dimensionality of the average scores of the 30 phrase groups, resulting in a component matrix and specific principal components. In the component matrix, variables with a high proportion of variance explained by the original data were selected, typically requiring eigenvalues greater than one and a cumulative variance contribution rate greater than 80%. Based on this criterion, three principal components were extracted, with a cumulative variance contribution rate of 96.002%. This indicates that components 1–3 encompass most of the original data's information, thus these three were identified as the main components, further validating the empirical and rational nature of the indicator design. The Principal Component Analysis Scale is shown in Table 6.

Table 5. Average value of the imagery phrases for the electric forklifts.

Number	Imagery Phrases	Average Value of Sample Scores			
		Option A	Option B	Option C	Option D
1	Monotonous–Creative	−0.286	−0.786	−0.786	1.357
2	Dynamic–Stable	−0.571	−1.000	−0.429	1.500
3	Harmonized–Dissonant	0.786	0.286	0.214	1.357
4	Comfortable–Uncomfortable	1.214	−0.429	0.786	1.500
5	Lightweight–Heavy	−0.357	−0.429	−1.286	1.357
6	Tough–Soft	0.786	1.214	1.071	0.214
7	Clean–Dark	1.071	−0.429	0.214	1.643
8	Low-profile–High-profile	0.143	0.214	0.643	0.786
9	Technological–Traditional	0.286	−1.143	−0.143	1.643
10	Beautiful–Ugly	0.643	−0.071	0.143	1.286
11	Simple–Complex	1.000	−0.786	0.429	1.500
12	Strong–Weak	0.857	1.286	1.214	0.429
13	Organized–Disorganized	0.500	0.786	0.214	0.500
14	Bright–Gray	1.214	−0.571	−0.071	1.643
15	Superior–Inferior	−0.143	−0.786	−0.429	1.143
16	Integral–Fragmented	0.929	0.714	1.000	1.286
17	Practical–Useless	0.714	0.857	0.714	0.929
18	Symmetrical–Arbitrary	0.571	0.929	0.714	0.571
19	Safe–Dangerous	0.857	1.357	0.857	0.857
20	Modern–Old-fashioned	0.214	−0.786	−0.429	1.214
21	Fine–Rough	0.571	0.714	0.143	1.286
22	Durable–Fragile	0.714	1.214	0.786	0.571
23	Exaggerated–Plain	−0.214	−1.000	0.071	1.643
24	Energy efficient–Energy consuming	−0.214	−0.357	−0.286	1.214
25	Intelligent–Manual	0.143	−0.714	−0.214	1.357
26	Conservative–Innovative	0.286	−0.857	0.143	1.714
27	Metal–Plastic	1.143	1.214	1.143	0.857
28	Expensive–Inexpensive	0.500	−0.143	0.857	1.214
29	Efficient–Inefficient	0.071	0.429	0.071	1.714
30	Reasonable–Unreasonable	1.286	1.286	0.929	1.357

Table 6. Principal Component Analysis (PCA) Scale.

Num	Principal Component Analysis		
	Eigenvalue	Variance Explained Ratio %	Cumulative %
1	12.875	82.603	82.603
2	1.396	8.958	91.561
3	0.692	4.441	96.002

Finally, the component matrix for the 30 sets of imagery phrases for the electric forklift is shown in Table 7. Components 1–3 captured most of the original data's information, and therefore, these three components were identified as the main components. From the component matrix in Table 5, the specific component loadings of each variable for components 1–3 could be derived. A higher loading indicated that the component had a stronger explanatory power for the variable. In component 1, 12 variables had higher loadings: variables 2, 3, 4, 5, 6, 11, 14, 16, 23, 25, 26, and 29. In component 2, two variables had higher loadings. In component 3, variable seven had a higher loading. Therefore, these 15 variables were determined to be the representative image words for the electric forklift.

Table 7. Imagery phrase component matrix for the electric forklifts.

Number	Imagery Phrases	Principal Component Analysis		
		Component 1	Component 2	Component 3
1	Monotonous–Creative	0.772	0.169	0.129
2	Dynamic–Stable	0.954	−0.014	0.299
3	Harmonized–Dissonant	0.961	0.239	−0.136
4	Comfortable–Uncomfortable	0.842	−0.443	−0.308
5	Lightweight–Heavy	0.860	0.496	0.116
6	Tough–Soft	−0.997	−0.060	0.024
7	Clean–Dark	0.644	−0.322	−0.804
8	Low-profile–High-profile	0.131	0.704	−0.407
9	Technological–Traditional	−0.337	0.929	−0.154
10	Beautiful–Ugly	0.198	−0.707	0.679
11	Simple–Complex	0.896	−0.350	−0.391
12	Strong–Weak	−0.786	−0.112	0.125
13	Organized–Disorganized	−0.660	−0.611	0.437
14	Bright–Gray	0.920	−0.035	−0.391
15	Superior–Inferior	0.789	0.401	0.221
16	Integral–Fragmented	0.930	−0.314	0.191
17	Practical–Useless	0.496	0.675	0.547
18	Symmetrical–Arbitrary	−0.740	0.392	0.455
19	Safe–Dangerous	−0.630	0.721	0.288
20	Modern–Old-fashioned	0.796	0.022	−0.092
21	Fine–Rough	0.740	0.655	0.151
22	Durable–Fragile	−0.647	0.487	0.214
23	Exaggerated–Plain	0.851	−0.795	0.251
24	Energy efficient–Energy consuming	0.743	0.158	0.293
25	Intelligent–Manual	0.889	0.400	0.221
26	Conservative–Innovative	0.978	−0.199	0.060
27	Metal–Plastic	−0.766	−0.14	−0.256
28	Expensive–Inexpensive	0.716	−0.552	0.169
29	Efficient–Inefficient	0.818	0.370	0.440
30	Reasonable–Unreasonable	0.460	0.848	−0.263

4.2. FKE Design Evaluation Process

The constructed design hierarchy model for an electric forklift is shown in Figure 5. The target layer is an electric forklift design evaluation, which is indicated by the letter Z. The 15 electric forklift phrases screened are used as the evaluation design elements in the second-level criterion layer, and are categorized into four features forming the first-level criterion layer, labelled "A form", "B color", "C style", and "D function".

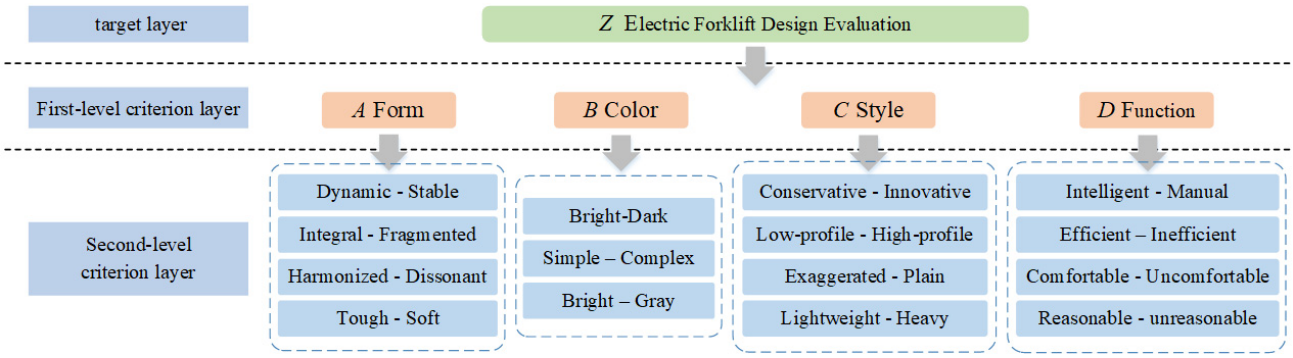


Figure 5. Design hierarchy model of an electric forklift.

In the survey of the criterion layer, considering that the format of the judgment matrix was difficult for the participants to understand, and despite detailed explanations, the results were still not ideal; thus, we chose to use a scoring method instead. A total of 20 questionnaires on the importance of electric forklift design indicators (Questionnaire S2 in Supplementary Materials) were distributed, and the research subjects came from teachers and graduate students of industrial design and vehicle engineering disciplines. In this study, all participants were fully informed of the research objectives and procedures before their involvement and provided their informed consent. We ensured the privacy of participants by anonymizing the data and maintaining strict confidentiality. Furthermore, participation in both the interviews and surveys was entirely voluntary, with participants being free to withdraw at any point without any consequences. The following was an example of the scoring process for the first indicator "Dynamic vs. Steady". The questionnaire question was: "How important do you think the 'Dynamic vs. Steady' indicator is in the design of electric forklift equipment? The scoring range is from 1 to 5, with the most important receiving a score of 5 and the least important receiving a score of 1. The scores decrease incrementally based on importance. Please give your score". Based on the statistical results of the questionnaire survey, we counted the number of people who assigned scores to the importance of each level and input the survey results into the AHP model. The statistical results of the research questionnaire were used to calculate the weight of each criterion layer, and each level of the judgment matrices and the weight results are shown in Tables 8–12.

Table 8. Judgment matrix and weight for electric forklift design.

	A	**B**	**C**	**D**	**Weight**
A	1.0000	3.0000	0.2500	0.5000	0.1504
B	0.3333	1.0000	0.2000	0.2500	0.0691
C	4.0000	5.0000	1.0000	3.0000	0.5349
D	2.0000	4.0000	0.3333	1.0000	0.2456

Table 9. Judgment matrix and weight of forms.

	A1	**A2**	**A3**	**A4**	**Weight**
A1	1.0000	2.0000	4.0000	3.0000	0.4668
A2	0.5000	1.0000	3.0000	2.0000	0.2776
A3	0.2500	0.3333	1.0000	0.5000	0.0953
A4	0.3333	0.5000	2.0000	1.0000	0.1603

Table 10. Judgment matrix and weight of colors.

	B1	**B2**	**B3**	**Weight**
B1	1.0000	0.5000	3.0000	0.3325
B2	2.0000	1.0000	3.0000	0.5278
B3	0.3333	0.3333	1.0000	0.1396

Table 11. Judgment matrix and weight of styles.

	C1	**C2**	**C3**	**C4**	**Weight**
C1	1.0000	5.0000	2.0000	4.0000	0.4851
C2	0.2000	1.0000	0.2000	0.5000	0.0725
C3	0.5000	5.0000	1.0000	3.0000	0.3192
C4	0.2500	2.0000	0.3333	1.0000	0.1232

Table 12. Judgment matrix and weight of functions.

	D1	D2	D3	D4	Weight
D1	1.0000	2.0000	0.3333	3.0000	0.2375
D2	0.5000	1.0000	0.3333	3.0000	0.1680
D3	3.0000	3.0000	1.0000	5.0000	0.5174
D4	0.3333	0.3333	0.2000	1.0000	0.0771

The calculation results show that all judgment matrices passed the consistency test. By normalizing the weight values of the criterion layer, the comprehensive weights of the judgment matrices of the second-level criterion layer were obtained, as shown in Table 13.

Table 13. Comprehensive weights and consistency tests of the secondary criterion layer.

First-Level Criterion Layer	Second-Level Criterion Layer	Single-Level Weights	Comprehensive Weights	Consistency Tests
A	A1	0.4668	0.1167	CR = 0.0116, CR < 0.1
	A2	0.2776	0.0694	
	A3	0.0953	0.0238	
	A4	0.1603	0.0401	
B	B1	0.3325	0.0831	CR = 0.0516, CR < 0.1
	B2	0.5278	0.1330	
	B3	0.1396	0.0349	
C	C1	0.4851	0.1213	CR = 0.0212, CR < 0.1
	C2	0.0725	0.0181	
	C3	0.3192	0.0798	
	C4	0.1232	0.0308	
D	D1	0.2375	0.0594	CR = 0.0390, CR < 0.1
	D2	0.1680	0.0420	
	D3	0.5174	0.1230	
	D4	0.0771	0.0193	

A comprehensive fuzzy evaluation of the electric forklift design was performed according to the indicator weight values. Given the objectivity of the calculation, it is necessary to quantify the evaluation results of the evaluation indicators. Five evaluation levels were determined—respectively, very good, good, general, bad, and very bad—and an assignment value for each level was provided. The assignment vectors are in the order $\beta = (90\ 80\ 70\ 60\ 50)^{T}$, and the evaluation levels and assignments are shown in Table 14.

Table 14. Evaluation levels and assignments.

Evaluation Levels	Very Good	Good	General	Bad	Very Bad
Assignments	90	80	70	60	50

Five industry experts and five designers were invited to evaluate the electric forklift design program. The evaluator scores were assigned according to the evaluation level standard and represent the total number of evaluation level assignments received by each evaluation index. For example, if evaluation index A1 received four "very good" evaluations, then the rating would be counted as 0.4 and the FCE matrix would be expressed as R1. The final evaluation matrices were obtained, respectively.

After obtaining the final evaluation matrices separately, the weighting results of option A were calculated according to Formula (7), and the second-level comprehensive evaluation matrix was established as follows:

$$P = \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \end{bmatrix} = \begin{bmatrix} 0.3371 & 0.2778 & 0.2818 & 0.1533 & 0.0000 \\ 0.1860 & 0.2528 & 0.2332 & 0.3139 & 0.0134 \\ 0.4464 & 0.2413 & 0.1907 & 0.1145 & 0.0072 \\ 0.4517 & 0.2643 & 0.1763 & 0.1077 & 0.0000 \end{bmatrix}$$

The results were converted to percentages to obtain the final score for Option A:

$$S_A = \begin{pmatrix} 0.4133 & 0.2532 & 0.2038 & 0.1324 & 0.0048 \end{pmatrix} \begin{pmatrix} 90 & 80 & 70 & 60 & 50 \end{pmatrix}^T = 79.67$$

The three remaining options were evaluated, respectively, according to this method to obtain a final score for each option:

The final score for Option B is $S_B = 72.79$.

The final score for Option C is $S_C = 80.05$.

The final score for Option D is $S_D = 80.60$.

The final score ranking is Option D > Option C > Option A > Option B. Accordingly, it can be determined that the evaluation result for Option D indicates the best electric forklift design.

4.3. BKE Validation of Design Evaluation

An ET experiment was used to validate the design evaluation via BKE. Evaluators were recruited from students majoring in industrial design, aged between 21 and 29. A total of 10 subjects (five male, five female) participated in a preliminary experiment. Participation in the experiment was completely voluntary, and subjects had the right to withdraw from the study at any time. We also anonymized the data to strictly protect the privacy of the participants, thus ensuring the confidentiality and anonymity of all participants. The experiment was conducted in the Experimental Building of the School of Design Art and Media, Nanjing University of Science and Technology, which is a quiet and comfortable place, ensuring that the experiment would not be disturbed by external noise. ET signals were recorded with a Tobii X3-120 eye tracker (Stockholm, Sweden), and the data were exported and analyzed using Tobii Studio.

Before the formal experiment began, the experimental procedure and precautions were explained to each subject. When the subjects had fully mastered the experimental procedure, the main experimenter explained the experimental instructions. In each trial, there was a period of time allowed for subjects to prepare and concentrate before designing the image presentation. During the experiment, the subjects were seated about 60 cm away from the eye tracker and maintained an upright posture.

The subjects were tested for visual selective attention according to the presented electric forklift schemes. Attention time and visual attention hotspots within the AOI were recorded in real time for each subject in turn for the four electric forklift design scenarios. After completing the experiment, eye movement type (fixation, saccade, blink), gaze tracking, and an indication of which AOI the gaze point was located in were exported. A case of one subject's area of interest is shown in Figure 6.



Figure 6. A case of one subject's AOI.

The average value of the time of attention within the AOI of the 10 designers was normalized to the data. Data were calculated and analyzed using the SPSS 25.0 software package (IBM, Armonk, NY, USA). The subjects were assessed for satisfaction with design preferences, and the evaluation dimensions were consistent with the vocabulary of the second-level criterion layer of PKE. The average values of the satisfaction scores with respect to electric forklift design preference are shown in Table 15.

Table 15. Average values of the satisfaction scores.

Phrases of Second-Level Criterion Layer	Option A	Option B	Option C	Option D
A1	−0.5	−1.2	−0.4	1.5
A2	0.9	0.7	1.2	1.3
A3	0.7	0.3	0.2	1.3
A4	0.8	1.2	1.2	0.2
B1	1.2	−0.4	0.2	1.6
B2	1.0	−0.8	0.4	1.5
B3	1.2	−0.5	−0.1	1.6
C1	−0.3	−0.8	−0.6	1.7
C2	−0.2	−1.0	0.2	1.6
C3	0.2	−1.1	−0.1	1.6
C4	−0.3	−0.7	−1.2	1.4
D1	0.4	−0.7	−0.2	1.3
D2	0.1	0.4	0.2	1.7
D3	1.2	−0.4	0.7	1.6
D4	1.3	1.2	0.9	1.3

According to Formulas (9) and (10), the satisfaction mean was dimensionless, and the absolute difference was solved. The gray correlation of each option was calculated as follows:

$$x_0 = (0.415, 0.373, 0.488, 0.863)$$

The results of the gray correlation calculation resulted in a final score ranking of Option D > Option C > Option A > Option B, thus determining that Option D was the best electric forklift design option. Consistent with the results of the PKE evaluation, this proves that the HKE model has a high degree of reliability, and the HKE model of this study is therefore determined to be valid.

To scientifically prove the validity of the subjective and objective evaluation method results, the root mean square error (RMSE) was applied to calculate the attention time and gray correlation within the interest area of each program. Using Formula (11), RMSE = 0.243 is obtained, which indicates that the program evaluation data based on BKE are reliable, and the subjective and objective evaluation methods have a certain degree of rationality.

To further validate the effectiveness of HKE, this study conducted an experimental comparison between the traditional KE approach and HKE. In traditional KE, the assessment mainly relies on subjective feedback from users and evaluation opinions from experts,

and the selection of design solutions is based on KE evaluation and qualitative analysis. Therefore, this study uses the results of user needs research and expert scoring as the basis of the traditional KE evaluation method. The scoring indicators are based on the 15 indicators screened in Section 4.2. In addition, in order to eliminate the influence of experts' personal subjective factors on the evaluation results, all the scorers in this comparative experiment are the members who actually participate in the scoring in this study, and the scoring scale is 1–10. The experimental process first invited five industry experts from the Section 4.2 FKE stage to score, then invited 10 designers from the Section 4.3 part as subjects to score. A comparison of the results of KE and HKE is shown in Table 16. The group of experts indicated in the interview stage after the scoring that the reason for choosing Option C as the best solution could be due to the fact that they instinctively considered the company's branding strategy and the products that they focused on promoting and selling, and did not focus their attention on the product design itself. The designer group scored similarly to the HKE option, but they indicated that samples with less variance in appearance and design tended to lead to cognitive fatigue. This also explains why the smaller differences in the ratings of the designer group failed to clearly reflect the differences between the schemes. In addition, the results showed that the BKE scheme compared better than FKE, while the similarity between the results of FKE and the designer group was higher, mainly because the source of data obtained from the AHP and FCE methods used by FCE was also the designer group's ratings.

Table 16. Comparison of results of KE and HKE.

KE Modeling	Comparative Method	Score For Each Option	Ranking of Option
KE	experts	$S_A = 39$; $S_B = 38$; $S_C = 43$; $S_D = 41$	C > D > A > B
	designers	$S_A = 79$; $S_B = 77$; $S_C = 80$; $S_D = 83$	D > C > A > B
HKE	FKE	$S_A = 79.67$; $S_B = 72.79$; $S_C = 80.05$; $S_D = 80.60$	D > C > A > B
	BKE	$S_A = 0.415$; $S_B = 0.373$; $S_C = 0.488$; $S_D = 0.863$	D > C > A > B

5. Discussion

In this study, the HKE model theory and AHP, FCE, ET, and GRA research methods were used to explore the evaluation of complex product designs using electric forklifts as a research case. Firstly, electric forklift designs and perceptual imagery vocabulary were collected and organized into 30 groups of perceptual imagery phrases after an initial screening. A PCA was then used for dimensionality reduction to obtain 15 representative imagery words that could be used to identify electric forklifts. The HKE model theory was then applied: The representative imagery words for the selected electric forklifts were substituted into the criterion layer to construct the hierarchical evaluation model of a forklift; the weights of the evaluation indexes of each layer and their ranking were calculated; and the ranking of the four electric forklift design solutions was calculated via the FCE method. An ET experiment was then performed to obtain attention duration and visual attention hotspots within the subjects' AOI of the forklift solutions, and the mean value of attention duration was normalized. The same representative intention words as those in FKE were used to evaluate the design preferences of the schemes, the gray correlation of each scheme was calculated, and the ranking of each design scheme was obtained. Finally, comparing the evaluation results of FKE and BKE thus proves the validity of this study. The proposed method in this study guides the design evaluation of electric forklift solutions, which in turn proves that the HKE model can effectively solve the problems found in complex product design evaluation.

The application of KE in product design is well established. Traditional KE has the advantage of simplicity and ease of use with low realization requirements; however, due

to the improvement in user requirements, it is gradually proving more difficult to employ traditional KE in effectively guiding certain types of product design innovations and evaluations. Some scholars have therefore conducted innovative and in-depth research on KE. Xing Liu et al. [19] proposed a product design method based on VR and KE, demonstrating that VR is more suitable for measuring and predicting users' reactions to future products. Yiru Jiao et al. [48] proposed a computerized method for the automatic extraction of KE that accelerates the traditional user survey process and specifies the emotional needs of users. Xinhui Kang [25] established a model based on KE, RST, and SVR and obtained the mapping relationship between users' KE and morphological features of the product, thus accurately grasping users' emotional cognition. Shen et al. [14] used fuzzy Kansei Engineering and creativity to conduct emotional product form research and designed new product prototypes that match customer preferences. Compared with previous studies, this study proposed research on an HKE-based design evaluation method that effectively improves the accuracy of product design evaluation while also providing a generalized solution for complex products.

The reduction in subjectivity is reflected in two main points: quantitative evaluation in FKE and objective validation in BKE. Through the use of AHP and FCE, users' perceived needs are transformed into quantifiable metrics, minimizing the subjective bias that often occurs in traditional expert ratings. Eye-tracking technology and GRA provide empirical data about users' preferences, ensuring that the assessment results have a scientific basis. While it is challenging to accurately calculate the percentage reduction in subjectivity, the dual validation framework ensures a higher degree of objectivity than traditional e-assessments.

The research objective of this study is complex product design, and the chosen case study is the electric forklift. Nevertheless, the proposed methodology in this study is generalizable and can be applied in studies on related product design. This study takes complex products as the research objective as the proposed method needs more steps and a longer time for completion and may not deliver efficiency advantages when applied to the evaluation of daily product design; however, this does not mean it cannot be used in that context, and the HKE-based design evaluation of complex products is more reliable than conventional evaluation methods in terms of accuracy. In addition, in this study, the AHP-FCE method was used for FKE, and the ET experiment evaluation method was used for reverse perceptual engineering. When design evaluation studies are conducted for other products, a new evaluation system based on HKE can be constructed by changing the evaluation methods, and the specific methods used for FKE and BKE should be determined according to the specific conditions of the products being designed. It is worth noting that the aim of design evaluation research based on KE is to increase accuracy, so its bi-directionality should not be verified via the evaluation methods of equivalent or similar methods, or those containing relationships among methods, such as FKE using AHP and BKE using FAHP, which cannot be called HKE and also provide little value.

Furthermore, we have found that this requires additional costs and time in practice. For example, the equipment and controlled environment required for eye-tracking experiments and additional data analysis methods such as GRA and RMSE all add to the complexity of the process. However, the advantages of the HKE dual validation process in providing a more reliable and accurate design assessment are clear. Especially for complex products such as electric forklifts, where design accuracy and user satisfaction are critical, the design phase plays a key role in the success of subsequent series production. Therefore, an upfront investment is necessary and justified. With HKE, we aim to achieve more accurate design evaluation, thereby reducing the likelihood of errors, redesigns, and user dissatisfaction later in the product lifecycle. While traditional KE or other design methods may appear to

be less resource intensive, they lack the robust validation provided by HKE and can lead to increased costs and delays later in the process. The added time and cost in the design phase is a strategic investment for mass production and higher user satisfaction later in the process, ensuring that the final product better meets user expectations and functions in the marketplace.

6. Conclusions

This study takes users' perceptual requirements as the design goal and proposes HKE-based complex product design evaluation research. Taking four electric forklift designs as an example, the evaluation indexes of forklift product styling imagery are obtained through the FKE model, and evaluation index systems of electric forklift styling imagery are constructed. The weights and rankings of evaluation indexes are calculated at each level using hierarchical analysis and FCE, which realizes the quantitative processing of each evaluation index in the evaluation index system. The reverse perceptual engineering model, combined with ET and GRA, obtains the preferred scheme according to the root mean square error calculation results and verifies the rationality and effectiveness of the theoretical model.

The method proposed in this study can help designers to better evaluate problems encountered in complex product design. Complex products are difficult to design and evaluate due to their high technical production requirements and numerous design indexes, and the complex product design evaluation method proposed herein is comprehensive, accurate, and effective. In addition, since the HKE modeling method contains multiple stages, both the FKE and BKE evaluation processes are compatible with multiple evaluation methods, which avoids the problem of a single evaluation perspective encountered when selecting a traditional method. This method improves the accuracy and comprehensiveness of design evaluation by designing and verifying the proposed two-way model, which in turn improves the accuracy of complex product design evaluation.

Although this study proposed HKE to improve the evaluation of complex product designs, it still has some limitations that deserve future exploration. Compared to KE, HKE is more complex both in the design process and in the combination of multiple methods, which also implies that more resources and time are required, especially when conducting eye-tracking experiments and analyzing data. Future research should focus on expanding the sample size to include more stakeholders, such as industry practitioners and potential users, with respect to the feasibility of time and resources, which will help to obtain broader feedback and more accurate evaluation results. Meanwhile, although this study provided valid conclusions using electric forklifts as a case study, it may not reflect the diversity of complex product categories. It is necessary for future research to explore a wider range of study subjects and to conduct further validation of other complex product categories in order to confirm the applicability and generalizability of the method. Additionally, this study mainly used PCA, AHP, FCE, ET, and GRA methods to validate the effectiveness of HKE; future work could further explore the incorporation of more diverse evaluation methods in HKE modeling, including other multi-criteria decision evaluation methods such as TOPSIS, structural equation modeling or best–worst method, and incorporate other physiological experimental evaluation methods, such as EMG experiments or EEG experiments. This allows for a more exhaustive comparison of different evaluation techniques and methods, thereby improving the robustness of HKE modeling.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/sym17020306/s1, Table S1. Initial perceptual imagery vocabulary library; Questionnaire S1: The questionnaire for electric forklift program design; Questionnaire S2: The questionnaire on the importance of electric forklift design indicators.

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Nomenclature

KE: Kansei Engineering; HKE: Hybrid Kansei Engineering; AHP: Analytic Hierarchy Process; FCE: Fuzzy Comprehensive Evaluation; ET: eye tracking; GRA: Gray Relational Analysis; RMSE: root mean square error; PF-AHP: Pythagorean fuzzy Analytic Hierarchy Process; RST: rough set theory; SVR: support vector regression; DSM: design structure matrix; EW: entropy weight method; WSM: Weighted Sum Model; MCDM: Multi-Criteria Decision-Making; TOPSIS: Technique for Order Preference by Similarity to an Ideal Solution; STNN: spatial–temporal neural network; EEG: electroencephalography; QFD: Quality Function Deployment.

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